

Employment and Education

UC Berkeley / Lawrence Berkeley National Lab Postdoctoral Researcher	from 2024
Stanford University PhD in Physics (advisor: Natalia Toro)	2024
Oxford University (New College) MSc in Mathematical and Theoretical Physics with distinction	2019
Cambridge University (St. John's College) MASt in Mathematics with distinction	2018
Massachusetts Institute of Technology BS in Physics and Mathematics	2017

Publications

Most of these works have alphabetical author lists; exceptions are indicated by underlining.

- 2609.10671** Ultralight Axial Dark Matter
A. Hook, J. Huang, K. Zhou
- 2607.27313** Suppressed Quantum Effects of Weakly Coupled Waves
Y. Bao, D. Y. Cheong, N. Rodd, J. Takach, L.-T. Wang, K. Zhou
- 2511.07514** Quantum Calculations of the Cavity Shift in Electron Magnetic Moment Measurements
F. Day, R. Harnik, Y. Kahn, S. Pavaskar, K. Zhou, JHEP 03, 038 (2026)
- 2510.05198** Intrinsically Quantum Effects of Axion Dark Matter are Undetectable
Y. Bao, D. Y. Cheong, N. Rodd, J. Takach, L.-T. Wang, K. Zhou,
Phys. Rev. Lett. 136, 171601 (2026)
- 2507.07173** A Prototype Hybrid Mode Cavity for Heterodyne Axion Detection
Z. Li, K. Zhou, M. Oriunno, A. Berlin, S. Calatroni, R. T. D'Agnolo,
S. A. R. Ellis, P. Schuster, S. Tantawi, N. Toro, NIMA 171474 (2026)
- 2504.02927** Determining Spin-Dependent Light Dark Matter Rates from Neutron Scattering
A. Berlin, A. J. Millar, T. Trickle, K. Zhou, Phys. Rev. D 112, 035021 (2025)
- 2502.01725** Ponderomotive Effects of Ultralight Dark Matter
K. Zhou, JHEP 05, 134 (2025)
- 2312.11601** Physical Signatures of Fermion-Coupled Axion Dark Matter
A. Berlin, A. J. Millar, T. Trickle, K. Zhou, JHEP 05, 314 (2024)

- 2303.04816** Interactions of Particles with “Continuous Spin” Fields
P. Schuster, N. Toro, K. Zhou, JHEP 04, 010 (2023)
- 2209.12901** Discovering QCD-Coupled Axion Dark Matter with Polarization Haloscopes
A. Berlin, K. Zhou, Phys. Rev. D 108, 035038 (2023)
- 2112.02104** Probing Invisible Vector Meson Decays with the NA64 and LDMX Experiments
P. Schuster, N. Toro, K. Zhou, Phys. Rev. D 105, 035036 (2022)
- 2106.09033** Stellar Shocks From Dark Matter Asteroid Impacts
A. Das, S. A. R. Ellis, P. Schuster, K. Zhou, Phys. Rev. Lett. 128, 021101 (2022)
- 2007.15656** Heterodyne Broadband Detection of Axion Dark Matter
A. Berlin, R. T. D’Agnolo, S. A. R. Ellis, K. Zhou, Phys. Rev. D 104, L111701 (2021)
- 1912.11048** Axion Dark Matter Detection by Superconducting Resonant Frequency Conversion
A. Berlin, R. T. D’Agnolo, S. A. R. Ellis, C. Nantista, J. Neilson,
P. Schuster, S. Tantawi, N. Toro, K. Zhou, JHEP 07, 088 (2020)
- 1704.06266** Casimir Meets Poisson: Improved Quark/Gluon Discrimination with Counting Observables
C. Frye, A. Larkoski, J. Thaler, K. Zhou, JHEP 09, 083 (2017)
- 1704.05456** Generalized Fragmentation Functions for Fractal Jet Observables
B. Elder, M. Procura, J. Thaler, W. Wallewijn, K. Zhou, JHEP 06, 085 (2017)
- 1703.04722** Minimum Energetic Cost to Maintain a Target Nonequilibrium State
J. Horowitz, [K. Zhou](#), J. England, Phys. Rev. E 95, 042102 (2017)

Community White Papers and Collaborations

- 2509.26574** Probing the Critical Point of AI Reasoning: a Frontier Physics Research Benchmark
Contributed a scaffolded and fully solved example research problem
- 2501.14249** Humanity’s Last Exam, Nature (2026)
Contributed tricky physics problems to help benchmark AI, interviewed in [New York Times](#)
- 2203.14923** Axion Dark Matter
Contributed writing for a subsection, and editing for all sections
- 2203.12714** Searches for New Particles, Dark Matter, and Gravitational Waves with SRF Cavities
Contributed writing and feedback
- 2203.08192** Current Status and Future Prospects for the Light Dark Matter eXperiment
Contributed theoretical projections for experimental sensitivity, and figures

Other Works

- [Physics Olympiad Handouts](#)
Proto-textbook with 1,000 tricky solved problems, used by students in dozens of countries

2411.08283 The surprising subtlety of electrostatic field lines
K. Zhou and T. Brauner, Am. J. Phys. 93, 234–240 (2025)

2203.15821 Comment on “Poynting vector controversy in axion modified electrodynamics”

Fellowships and Awards

NSF Graduate Research Fellowship	2017 – 2022
Marshall Scholarship	2017 – 2019
Demuth Prize, New College	2019
Dirac Prize, St. John’s College	2018
Finalist, Hertz Fellowship	2017
Joel Matthew Orloff Award for Outstanding Research, MIT	2017
Honorable Mention, Putnam Mathematical Competition	2016, 2017
Gold Medal, International Physics Olympiad	2012, 2013
Winner, USA Junior Mathematical Olympiad	2011

Grants

An SRF Cavity for Dark Matter Axion Detection	2022 – 2024
• SLAC LDRD grant, with principal investigator Zenghai Li • Participated in design discussions, writing and editing of proposal and final publication	

Seminars

Ultralight Axial Dark Matter Fermilab Particle Theory Seminar	8/26
Suppressed Quantum Effects of Weakly Coupled Waves KAIST Seminar	7/26
The Cavity Shift of Electron $g - 2$ Seoul National University Seminar	7/26
Korea Institute for Advanced Study (KIAS) Seminar	7/26
Southern University of Science and Technology Seminar	12/25
UC Davis QMAP Particle/Cosmology Seminar	11/25
Fudan University Seminar	7/25
Quantum and Classical Effects of Axion Dark Matter University of Toronto THEP Seminar	4/26
Cornell Particle Theory Seminar	4/26
Hong Kong City University Seminar	12/25
Hong Kong University of Science and Technology Seminar	12/25
Chinese University of Hong Kong Seminar	12/25
UC Berkeley “4D” Seminar	10/25
Nanjing Normal University Seminar	7/25
Tsung-Dao Lee Institute Seminar	7/25

New Precision Experiments for Axion Dark Matter	
Institute for Basic Science (IBS) DMAG Seminar	7/26
Tsinghua University Seminar	7/25

How (Not) to Probe the Axion-Electron Coupling

Peking University Seminar	7/25
UC Davis "Xperiment" Seminar	5/25
University of Chicago Particle Theory Seminar	4/25
UC Berkeley "4D" Seminar	8/24
Flatiron Institute, Particle Astrophysics and Cosmology Meeting Around NYC	4/24
University of Geneva High Energy Particle Physics Seminar	3/24
HEP/Astro Results Forum	3/24
SLAC Theory Seminar	11/23

Electromagnetism and Gravity with Continuous Spin

Tsinghua University Seminar	7/25
UIUC High Energy Physics Seminar	10/24
Hunting Invisibles (HIDDeN) Virtual Institute Seminar	11/23
Caltech High Energy Physics Seminar	10/23
UC Santa Cruz SCIPP Seminar	10/23
University of Maryland EPT Seminar	9/23
ICTP HECAP Seminar	7/23
CERN BSM Forum	6/23
UC Davis QMAP Particle/Cosmology Seminar	4/23
UC Berkeley "4D" Seminar	4/23
Stanford Phenomenology Seminar	2/23
Perimeter Institute Theory Seminar	10/22

Discovering the QCD Axion with Polarization Haloscopes

Fermilab Particle Theory Seminar	4/23
TRIUMF Theory Seminar	10/22
University of Victoria Theory Seminar	10/22

Flashes in the Dark: New Searches for Axions and Macroscopic Dark Matter

Johns Hopkins Theory Seminar	9/22
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Searching for Ultraheavy and Ultralight Dark Matter

SLAC Theory Seminar	3/22
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Heterodyne Detection of Axion Dark Matter

Virtual Axion Institute	8/20
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Conferences and Workshops

Ultralight Axial Dark Matter	
Dark Frontiers 2026 (invited)	8/26

The Superconducting Heterodyne Approach to Axion Detection

Axions in Stockholm 2025	7/25
Berkeley Axion Workshop 2025 (invited)	5/25

Spin-Dependent Dark Matter Rates from Neutron Scattering	
Phenomenology 2025 Symposium	5/25
Physical Signatures of Fermion-Coupled Axion Dark Matter	
Phenomenology 2024 Symposium	5/24
Discovering the QCD Axion with Polarization Haloscopes	
18th Patras Workshop on Axions, WIMPs and WISPs	7/23
Phenomenology 2023 Symposium	5/23
Probing Dark Sectors With Invisible Vector Meson Decays	
Phenomenology 2022 Symposium	5/22
APS April Meeting 2022	4/22
ILCX2021: ILC Workshop on Potential Experiments	10/21
Stellar Shocks From Dark Asteroids	
COSMO'21: 24th International Conference on Particle Physics and Cosmology	8/21
DPF21: APS Division of Particles & Fields Meeting	7/21
Phenomenology 2021 Symposium	5/21

Outreach and Service

U.S. Physics Olympiad	2015 – 2024
- Wrote and edited the largest physics competition in the United States (6,000 participants) - Taught classes on problem solving and lab skills to finalists at annual training camps - Managed the theoretical training of the U.S. traveling team from 2021 to 2024	
Press coverage	2022
Participated in several interviews for “Stellar Shocks From Dark Matter Asteroid Impacts” (Altmetric score of 200+, in top 1% of Physical Review Letters)	
Online discussion	
- Wrote 1,000 answers on StackExchange for all fields of physics (3 million total views) - Moderator of Reddit’s r/Physics, the internet’s most active physics forum (1 million views per week)	
National Science Bowl	2023, 2024
Wrote and edited physics questions for the U.S. Department of Energy’s flagship middle school and high school outreach event (~10,000 participants)	
Department service	
Co-organized the Berkeley “Axion Works” seminar	2025–2026
Co-organized the Berkeley/LBNL particle theory seminar	2025
Served as student representative for the physics department’s Graduate Studies Committee	2023
Participated on various panels for undergraduates and incoming graduate students	2020
Local outreach	
Mentored a local undergraduate research intern	Summer 2024
Judged research presentations for the US Invitational Young Physicists Tournament	2023
Taught high school students at “Splash” events at MIT, Oxford, and Stanford	2013 – 2019

Peer review

- Refereed research papers for Phys. Rev. Lett, Phys. Rev. D, Nature Communications, JHEP, JCAP, Phys. Lett. B, and Int. J. Mod. Phys. A
- Refereed pedagogical papers and books for Am. J. Phys, Cambridge University Press, and World Scientific

Teaching and Education

Physics 230: Graduate Quantum Mechanics I 2024

Physics 120: Intermediate Electricity and Magnetism I 2023

Physics 330: Quantum Field Theory I 2022

- Ran weekly sections and office hours; helped write, edit, solve, and grade [new problem sets](#)